

Week 1 · Laboratory Solutions

- These are **reference** answers, not the only correct ones. The checker tests the physics, not the diagram.
- Read them after attempting the problem. Each script explains what to build **and why the alternatives were rejected**, which is the part that does not survive being read backwards from a finished model.

Building and checking all three

```
cd lectures/W01_simulink/solutions
W01_S1_openloop      % builds W01_S1.slx
W01_S2_manoeuvre     % builds W01_S2.slx
W01_S3_current       % builds W01_S3.slx

W01_check(1,'W01_S1')
W01_check(2,'W01_S2')
W01_check(3,'W01_S3')
```

All three pass. The output below is the measured one, not a transcription.

```
W01 problem 1 - checking model W01_S1
terminal surge speed u          1.0286 (expected  1.0286 +- 0.005 m/s) PASS
sway velocity v (must stay zero) 0.0000 (expected  0.0000 +- 1e-06 m/s) PASS
yaw rate r (nothing steers)     0.0000 (expected  0.0000 +- 1e-06 deg/s) PASS

W01 problem 2 - checking model W01_S2
yaw rate in the PORT turn       -2.2952 (expected -2.2942 +- 0.02 deg/s) PASS
yaw rate in the STARBOARD turn  2.2955 (expected  2.2942 +- 0.02 deg/s) PASS
heading change, port turn       -70.1965 (expected -70.2000 +- 0.5 deg) PASS
heading change, starboard turn  70.6083 (expected  70.6000 +- 0.5 deg) PASS
sway v in the port turn         0.1263 (expected  0.1264 +- 0.005 m/s) PASS
sway v in the starboard turn    -0.1262 (expected -0.1264 +- 0.005 m/s) PASS

W01 problem 3 - checking model W01_S3
ground speed over the ground    1.1091 (expected  1.1091 +- 0.02 m/s) PASS
track direction from North      21.8041 (expected  21.8040 +- 0.3 deg) PASS
drift, track minus heading      25.8088 (expected  25.8090 +- 0.3 deg) PASS
```

What each solution is really teaching

	The block that matters	The idea it carries
S1	Constant [n0 ; n0]	The command is a 2-vector , and equal entries mean no turn, because $N = y_p(T_{left} - T_{right})$
S2	Reshape → 1-D array	A MATLAB Function writing <code>n = [nL; nR]</code> emits a 2 × 1 matrix ; the plant tolerates it and the log does not
S3	none — the canvas is unchanged	A current is a velocity , not a force. Forces use ν_r ; position uses ν

The one measurement trap worth knowing about

Problem 3 asks for the angle between the track and the heading. There are two reasonable ways to measure a "track direction", and they disagree by about half a degree on this run:

Definition	Value
direction of the straight line from start to finish	21.804°
instantaneous direction of the velocity, averaged over the settled fifth	22.366°

They differ because **the vessel weathervanes about 4° during the run**, so the velocity at the end is not parallel to the net displacement. Section E of the lecture uses the first definition, and so does `W01_check`.

This is not pedantry about conventions. It is the reason a measured number always has to be paired with the sentence that says what was measured — a rule this course applies to its own figures as well.